Project 1 — The Five Classes of Learners

Fundamentos de Aprendizagem Automática

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2026

1. Goal

Pedro Domingos (*The Master Algorithm*, 2015) groups machine learning into **five classes of learners**, each with its own idea of what learning is. You will pick **one learner from each class**, train the five on **one dataset**, and compare them with a methodology you define and defend. The question is not "which model won", but **why**: what each class assumes about the data, and what the plots and the numbers say about those assumptions.

Choose a task: **classification** or **regression**. Use the same task for all five learners.

2. The Five Classes of Learners

Class	Learns by	Typical learners (scikit-learn, Keras, pyBlindOpt)	Taught
Symbolists	inducing rules by inverse deduction	decision tree, rule list, extra-trees; boosted trees only if you justify them as symbolic	Class 06–07
Connectionists	adjusting the weights of a network by backpropagation	MLP (Keras with the JAX backend, or `MLPClassifier`/`MLPRegressor`), logistic regression as a one-neuron network	Class 03, 05

Class	Learns by	Typical learners (scikit-learn, Keras, pyBlindOpt)	Taught
Evolutionaries	evolving a population of candidate models	the *parameters*, the *features* or the *rules* of a model found by a genetic algorithm or differential evolution (`pyBlindOpt`), optimizing the metric directly	Class 02
Bayesians	updating probabilities with Bayes' rule	Gaussian or categorical Naive Bayes, Naive Bayes regression, Bayesian ridge, a model fitted by MAP	Class 04
Analogizers	comparing a new case with similar known cases	k-NN, support vector machine or SVR with a kernel	Class 05–06

Notes:

- One learner **per class**, five in total. Different settings of the same learner (two trees with different depths) count as one.
- Tuning a hyper-parameter with an evolutionary algorithm does **not** make a learner evolutionary: here the evolution must be the way the model is *learned*.
- A class may have borderline members (an ensemble, a regularized linear model). You may use one, but your README must say why it belongs to the class.
- Some classes are taught after the release (the last ones in Class 06–07): start with the ones you know.

3. What To Do

Follow these steps, **in this order**; they map to the sections of your README (Section 5).

1. **Choose the dataset** (Section 4): the task, the target, what a good model is worth in the real problem.
2. **Data visualization.** Look at the data before any model: the target (distribution, classes balance), the features (distributions, scales, outliers, missing

values), the relations (correlations, the features against the target). Every plot must lead to a decision.

3. **Evaluation methodology.** Define how you will compare the learners: the split (hold-out, stratified, k-fold, repeated), the metrics and why they match the problem, the baseline (the mean, the majority class), and how you quantify the spread so that a difference is not noise.
4. **Preprocessing.** Define what the data need (missing values, categorical encoding, scaling, transformations). Every statistic is learned on the training part only, **inside** each fold; each learner may need a different preprocessing, and you explain why.
5. **Models.** Fit and tune each learner on validation data only. Report the search space and the tuning cost, not only the best result.
6. **Evaluation and conclusion.** Compare the five learners under the same methodology; analyse errors and costs, and relate the outcome to what the data looked like in step 2.

Constraints:

- Use the course stack: **polars** for the data, **numpy** and **jax** for computation, **matplotlib** and **seaborn** for plots, **scikit-learn** and **keras** for the models, **pyBlindOpt** for the evolutionary learner.
- Everything must run in a few minutes on a laptop (the datasets below do). Fix every random seed.
- A test set (or the outer folds) is used **once**, at the end, for the final numbers.

4. Datasets

Use one of the suggestions, which are real, small and already in the repository (`datasets/`, as `.csv.zst` files that polars reads directly; sources and columns in `datasets/README.md`), or propose your own. A dataset of your own must be real, public, run on a laptop and be agreed beforehand with your instructor.

Dataset	Task	Size	What makes it interesting
`breast_cancer`	classification	569 × 30	clean and easy: a ceiling effect, strongly correlated features, the cost of a missed tumour

Dataset	Task	Size	What makes it interesting
`german_credit`	classification	1000 × 20	categorical and numeric features; wrong decisions have different costs (5 : 1)
`heart_disease`	classification	303 × 13	very small, categorical, **missing values**: the variance of the estimate is the problem
`spambase`	classification	4601 × 57	very skewed features, many zeros, duplicates, a threshold to choose
`digits`	classification, 10 classes	1797 × 64	multi-class; images as vectors; k-NN and SVM shine
`california_housing`	regression	20640 × 8	geographic structure, a capped target, skew (class 03); heavy for k-NN and SVR: subsample and say so
`concrete`	regression	1030 × 8	a smooth non-linear function with interactions
`abalone`	regression	4177 × 8	a categorical feature and a noisy target
`auto_mpg`	regression	398 × 7	tiny, with missing values, correlated features
`wine_quality`	regression (or classification)	6497 × 11	an integer target on a few levels, unbalanced
`diabetes`	regression	442 × 10	tiny and noisy; low ceiling for every learner

Justify your choice: why this dataset, what the five classes are *expected* to do on it (a hypothesis you can confirm or reject), and what its limits are.

5. Submission

A repository (or a zip) with:

1. **The code**: a Jupyter notebook **or** Python files, runnable from top to bottom with the course environment (`make venv`), reading the dataset from `datasets/`. Keep the notebook clean: no unused cells, outputs allowed.
2. **`README.md`** with *all* the information, with these sections:

Section	Content
1. Identification	names and student numbers, course, date, the dataset and the task in one line
2. Models	the five learners, one per class: what it is, why you chose it for this dataset, its hyper-parameters and search space
3. Dataset	the choice and its justification, the source, the size and features; the data visualization with its findings
4. Methodology	the evaluation methodology (split, folds, metrics, baseline, spread) and the preprocessing (steps, and where they are fitted); the tuning protocol; how the pipeline avoids leakage
5. Evaluation and conclusion	the results table (mean and spread, with the baseline), the plots, the analysis of errors, the answer to your hypotheses, the limits, and what you would do next

Plots and diagrams are mandatory, at least:

- a **diagram** of the pipeline (data, visualization, preprocessing, folds, tuning, test) with the five learners;
- the **data visualization** of Section 3 (target, features, relations);
- a **comparison plot** of the five learners with their spread (not only means);
- a **diagnostic plot** for the best learner (confusion matrix and threshold curve, or predicted against actual and residuals);
- a **learning or validation curve** for at least one learner.

Every figure has a caption and is discussed in the text. Save the figures in the repository and link them from the `README.md`.

Assessment. We look at: the quality of the justifications; the correctness of the methodology (no leakage, a baseline, a spread); the quality and the use of the plots; the five classes represented honestly; the conclusions supported by the numbers; reproducibility.

6. Dates

- **Release:** Class 03 (01/02 October 2026)
- **Submission deadline:** Class 08 (05/06 November 2026)
- **Weight:** 25% of the final grade